

# MINGYU KIM

## CONTACT INFORMATION

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**Email:** [k0136000@mju.ac.kr](mailto:k0136000@mju.ac.kr)

**Website:** <https://kminsalgorithm.github.io>

## RESEARCH INTEREST

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My master's research focused on enhancing dynamic obstacle avoidance performance in mobile robot navigation by leveraging deep learning models to process semantic information and generate human-like trajectories based on imitation learning.

In the future, my primary research interests will focus on:

- Reinforcement Learning for Robotic Manipulation
- Vision-Language-Action model
- Multimodal learning

## EDUCATIONAL QUALIFICATION

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**Myongji University**

M.S. in Applied Artificial Intelligence  
Total GPA of 4.5/4.5

Yongin, Korea  
Mar. 2024 - Current

**Myongji University**

B.S. in Information and Communication Engineering  
Total GPA of 3.5/4.5, Major GPA of 3.77/4.5

Yongin, Korea  
Mar. 2020 - Feb. 2024

## PUBLICATIONS

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### International Journals

- **HADP: Hybrid A\*-Diffusion Planner for Robust Navigation in Dynamic Obstacle Environments** [paper](#)  
M. Kim\*, C. Heo\*, J. Jung (\*co-first authors)  
*IEEE Access* [Impact Factor: 3.6]  
[overview](#)
- **HiMSELF: A Hierarchical Misbehavior Classification with Sequence Embedding by Latent Features in Vehicular Ad-Hoc Networks** [paper](#)  
M. Kim, D. Yum, J. Jung  
*IEEE Access* [Impact Factor: 3.6]  
[overview](#)

Domestic Journals

- **Effective Embedding Techniques for Misbehavior Classification in Vehicular Ad-Hoc Networks** [paper](#)  
M. Kim, J. Jung  
Journal of Korean Institute of Information Scientists and Engineers, (KIISE 2024).

Domestic Conferences

- **Label Similarity Analysis Based on LSTM in Vehicular Ad-hoc Networks** [paper](#)  
M. Kim, J. Jung  
Korea Software Congress, (KSC 2023) – **Best Paper Award**.
- **Reinforcement Learning-Based Navigation System Without Dependence on SLAM** [paper](#)  
M. Kim, J. Jung  
Korea Institute of Next Generation Computing, (KINGPC 2024).

RESEARCH PROJECTS

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|--------------------------------------------------------------------------------------------------------------------|-------------------|
| Development of a Cybersecurity Platform for V2X-based Remote-Assisted Mobility                                     | 2024.03 – Current |
| AutoCrypt                                                                                                          |                   |
| Semantic Autonomous Driving Systems Utilizing Diffusion Models for Enhanced Dynamic Obstacle Avoidance Performance |                   |
| National Research Foundation of Korea (NRF)                                                                        | 2024.07 – 2025.07 |

PATENTS

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|----------------------------------------------------------------------------------------|------------------------------------------------|
| Method for producing dynamic obstacle avoidance path using conditional diffusion model |                                                |
| Status:                                                                                | Patent Application Filed                       |
| Application No.:                                                                       | 10-2025-0053197                                |
| Filing Date:                                                                           | Apr 23, 2025                                   |
| Applicants:                                                                            | Jaehee Jung, <b>Mingyu Kim</b> , Chanyeong Heo |
| System for producing dynamic obstacle avoidance path using conditional diffusion model |                                                |
| Status:                                                                                | Patent Application Filed                       |
| Application No.:                                                                       | 10-2025-0053198                                |
| Filing Date:                                                                           | Apr 23, 2025                                   |
| Applicants:                                                                            | Jaehee Jung, <b>Mingyu Kim</b> , Chanyeong Heo |

AWARDS

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|                                                                                                            |      |
|------------------------------------------------------------------------------------------------------------|------|
| Encouragement Prize, Myongji University Software Competition                                               | 2022 |
| Grand Prize, Myongji University Software Competition                                                       | 2023 |
| Best Paper Award, Korea Software Congress (KSC)                                                            | 2023 |
| Gold Prize in the Contribution Award, Myongji University                                                   | 2024 |
| Awarded the Research Scholarship for M.S. Studies by the National Research Foundation (NRF) - \$12,000,000 | 2024 |

## SKILLS

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### English

- TOEIC: 770

### Programming Language

- Python (PyTorch, Tensorflow, FastAPI), C (MCU programming), JavaScript (NestJS)

### Machine Learning

- Graph Neural Network, Imitation learning, Anomaly Detection

### Robot Development and Implementation

- Learning-Based Path Planning
- Development of a [Swerve-Drive Robot](#)
- Development of an [Omni-Wheel-Based Robot](#)

### Development Tools

- ROS, Gazebo, Fusion 360

## EXTRACURRICULAR ACTIVITIES:

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**Undergraduate Research Intern, DS LAB (Advisor: [Jaehee Jung](#))** 2022 – 2024

- Researched on improving misbehavior classification in V2X messages by analyzing relationships between misbehavior types.
- Researched on DRL-based navigation systems that operate without reliance on maps.

**LikeLion, Myongji University** 2023 – 2024

- Backend Operator & Instructor

**AUTURBO** 2024 – 2025

- Member, Outdoor Delivery Robot Team

## PERSONAL INFORMATION:

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**Date of Birth** 20-02-2001

**Hobbies** 3D Modeling  
Freelance work via [Soomgo](#)